



Say it once. Every screen knows.

Built on the AssemblyAI Voice Agent API. A cook says “86 salmon” into a headset — the moment the transcript lands, the kitchen display, the servers’ screen and the online ordering page all agree.

When a kitchen runs out, the news travels by shouting — **and the apps keep selling it.**

≈75%

of US restaurant traffic is off-premises — the screens are the storefront.

NRA, OFF-PREMISES
TRENDS 2025

1.5–3%

of US online orders were cancelled in 2023 — and cancellations track the rating.

OTTER, 2023 ORDER
DATA

4.5★ → 4.0★

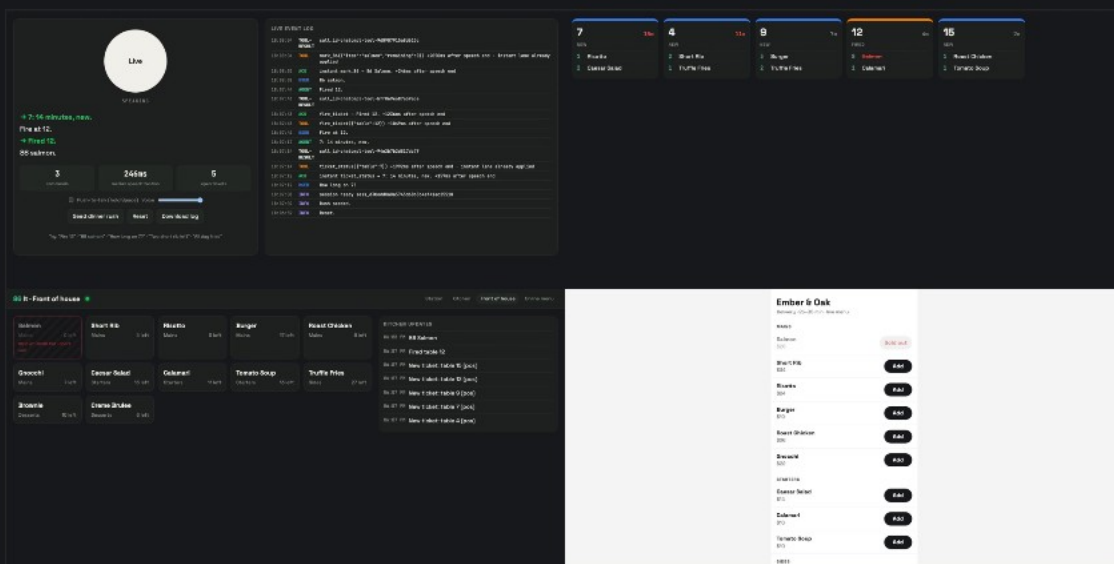
average rating below 2% cancellations vs above 8% — two in a row can pause the store.

OTTER, DECEMBER
2023

GREASY GLOVES CAN'T TAP SCREENS

TOO LOUD FOR A VOICE ASSISTANT

One sentence — “86 salmon” — and **every** screen changes in the same moment.



Kitchen display

Salmon greys out — the line stops cooking it.

Servers' screen

Front of house can no longer sell it.

Online menu

Delivery pages reject it the same instant.

ONE UTTERANCE → THREE SCREENS

0 MS INSTANT • 2.1 S LLM

Cooks speak kitchen slang; the system answers in **one to five words**.

86 salmon	Greys out on KDS, servers' screen, menu.
two short rib left	Count updates everywhere, item flagged low.
fire 12 · bump 4 · hold 9	The ticket moves on the kitchen display.
how long on 7?	"Seven: 14 minutes." · "All day: three."
12 has a nut allergy	A red banner appears on ticket 12.
void 7 → "yes"	"Void 7?" first — nothing until "yes".
kitchen chatter	Silence: no tool call, no spoken reply.
ticket passes 10 min	Speaks unprompted: "Table 4, 12 minutes."

ENUMS RESTRICT THE LLM TO MENU ITEMS KEYTERMS CARRY THE SLANG

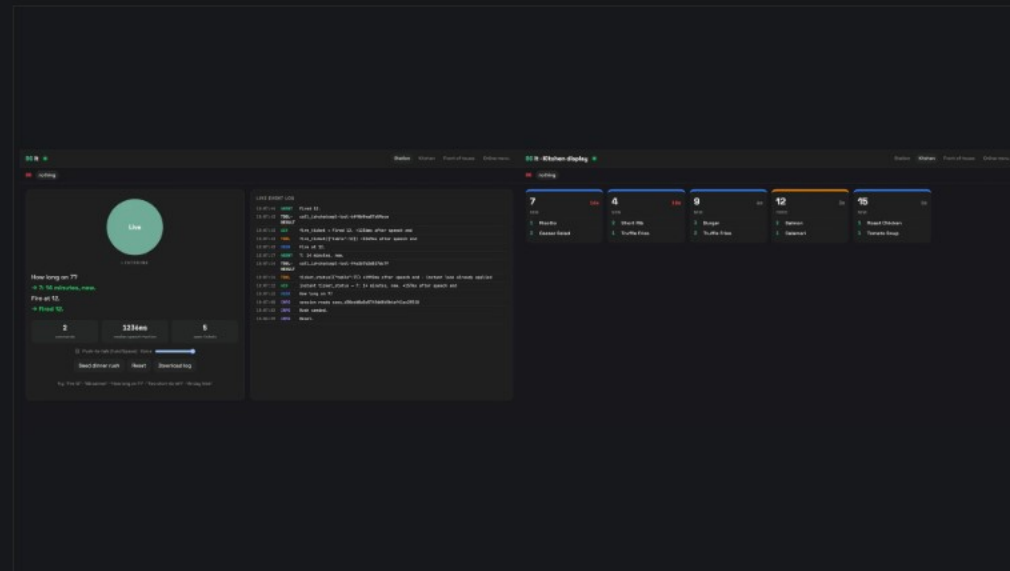
The same voice loop is **deployed and live** — this is what runs on stage.

0:20 “Fire twelve.” → ticket 12 turns orange.

0:40 “Eighty-six salmon.” → three screens grey out.

1:25 A teammate talks football — nothing happens.

1:55 Late-ticket alert: “Table 4, 12 minutes.”



URNS ORANGE MID-REPLY

One WebSocket carries the voice; **one Durable Object owns the state** and fans it out.

- 01 Station** — browser + headset: 24 kHz PCM16 streams.
↓ **INPUT.AUDIO, STREAMED AS THE COOK SPEAKS**
- 02 AssemblyAI Voice Agent** — STT, turn detection, LLM + 10 tools, TTS.
↓ **TRANSCRIPT → INSTANT LANE ACTS; LLM CALL DEDUPED**
- 03 Worker + Durable Object** — tokens, authoritative state, fan-out.
↓ **STATE BROADCAST TO EVERY OPEN SOCKET**
- 04 Every screen** — kitchen display, servers' board, online menu.

Screens change on the broadcast — **before** the agent speaks.

GRAMMAR FIRST • LLM CONFIRMS TEMP TOKEN, KEY STAYS SERVER-SIDE

Every hard part of voice is an AssemblyAI feature we wired up — **not code we wrote.**

VOICE AGENT API

One WebSocket: STT, LLM, speech, turn detection.

KEYTERMS + MENU ENUMS

Slang transcribes; enums stop invented dishes.

MIN_LATENCY + BARGE-IN

Fast transcription; the cook can cut the agent off.

TEMP TOKENS FROM /API/TOKEN

INLINE SESSION CONFIG

Prompt, 10 tools and keyterms in session.update.

VOICE FOCUS • NEAR-FIELD

Isolates the cook from a loud line, every time.

REPLY.CREATE + RESUME

Unprompted alerts; survives a Wi-Fi blip.

CLIENT TOOLS DECLARED INLINE

43 of 43 clips pass — 42 of 43 in noise.

43-CLIP EVAL	CLEAN	PINK NOISE -12 DB
Pass · LLM only	42 / 43	42 / 43
Pass · + instant lane	43 / 43	42 / 43
Chatter ignored	12 / 12	12 / 12
Speech → action (p50)	0 ms · LLM 2.1 s	0 ms · LLM 2.0 s
Spoken confirm (p50)	4.9 s	4.2 s

The one noisy miss was a mishearing — and it failed **safe**.

22 unit tests and a protocol-faithful fake caught violations before we spent a credit.

22

unit tests over the kitchen reducer, fuzzy matching, void safety and alerts.

```
NPM TEST • NODE --  
TEST
```

0

protocol violations across a full voice loop: tools, barge-in, chatter, resume.

```
GET :9901/STATS →  
VIOLATIONS: []
```

86

sessions recorded with AssemblyAI's own recording and timeline.

```
RECORDINGS/ •  
EVAL/RESULTS/
```

THE MOCK FLAGS PROTOCOL VIOLATIONS

ON EVERY RECORDED SESSION

Per-location subscriptions, sold through **the POS vendors restaurants already run.**

BUYER

The owner or kitchen manager, for independents and groups.

DISTRIBUTION

An add-on for POS/KDS vendors: Toast, Square, Lightspeed.

NEXT: MULTI-STATION KITCHENS • MULTI-LOCATION DASHBOARDS •
MANAGER ANALYTICS

PRICE

Per location per month, optional headset + tablet bundle.

VALUE

Fewer refunds, faster tickets, allergy flags, a timestamped 86 log.

DEMO POS IS A MOCK

ONE DURABLE OBJECT PER RESTAURANT

What is honestly **not finished yet.**

LIMITS TODAY

One restaurant per deployment; no auth.

The instant lane covers fixed phrasings only.

The POS is a mock.

Eval used synthesized speech.

Off-grammar speech takes 2–4 s;
“Un-86 X” is flaky.

NEXT

Key the Durable Object by restaurant id.

Add auth to the demo screens.

An owner phone line: “what did we 86 tonight?”

Real voices and noise in the eval.

Toast / Square inventory via act().

Say it once. Every screen knows.

Live `86-it.rd5510.workers.dev` **Write-up** `docs/SUBMISSION.md`

Video `86-it-demo-final.mp4` • 2:16

Sending tool results immediately **cut two seconds of dead air** from every confirmation.

WHAT THE DOCS SUGGESTED

Wait for `reply.done` before sending `tool.result` — the agent may be speaking a transition.

WHAT THE TIMELINES SHOWED

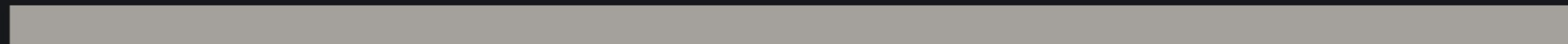
That transition is silent. We send `tool.result` immediately:
confirmations ~2 s sooner, zero errors in any recorded session.

MEASURED AND REVERTED: `EXECUTION_MODE HOLD` • `SHORTER PROMPT` •
`MIN_SILENCE 400 MS`

transcription_mode: min_latency **bought 0.6–1 s** — at the same accuracy.

BALANCED MODE • 3-COMMAND A/B

2.85 – 4.3 S



MIN_LATENCY • SAME 3 COMMANDS

2.2 – 3.7 S



Prompt fixes that landed: chatter gets silence, menu nicknames (“the fish” → salmon), number homophones, and a silent-reply gate.